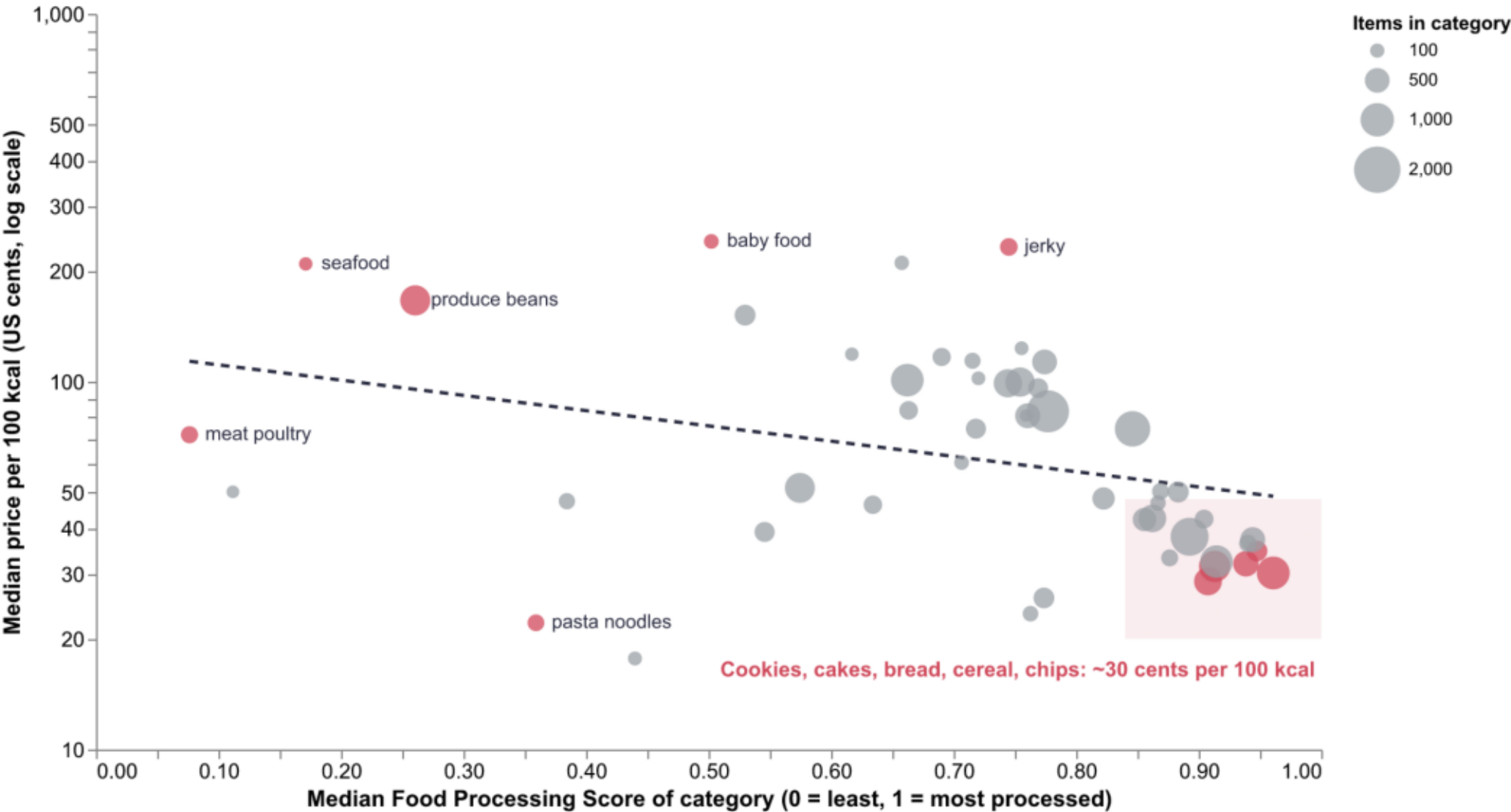


Plot A

The more processed the food, the cheaper its calories

Each dot is one of 50 food categories (categories with 20+ items priced per calorie; coffee beans excluded). Value = category median across Walmart, Target and Whole Foods items. Dashed line: log-linear fit.

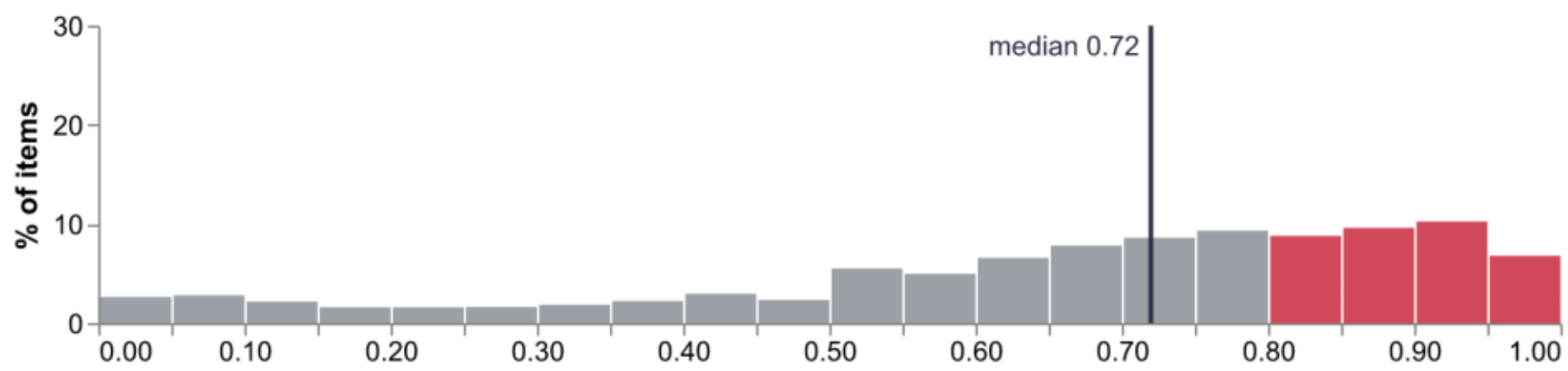


Plot B

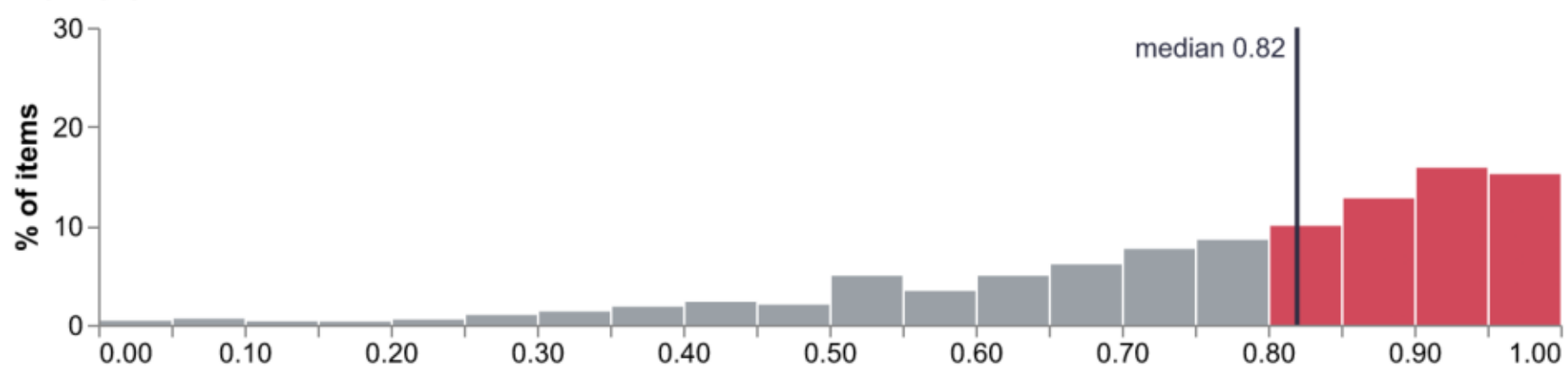
Whole Foods stocks far more minimally processed food than Walmart or Target

Histogram of Food Processing Score per store (20 bins, all 26,250 items); red = score 0.8 or higher (heavily processed).

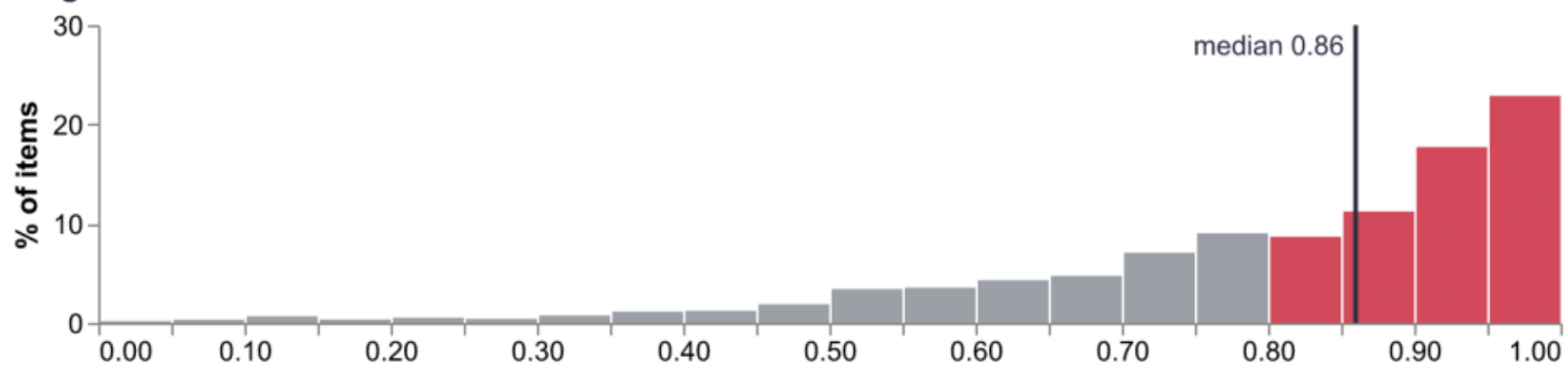
Whole Foods



Walmart



Target

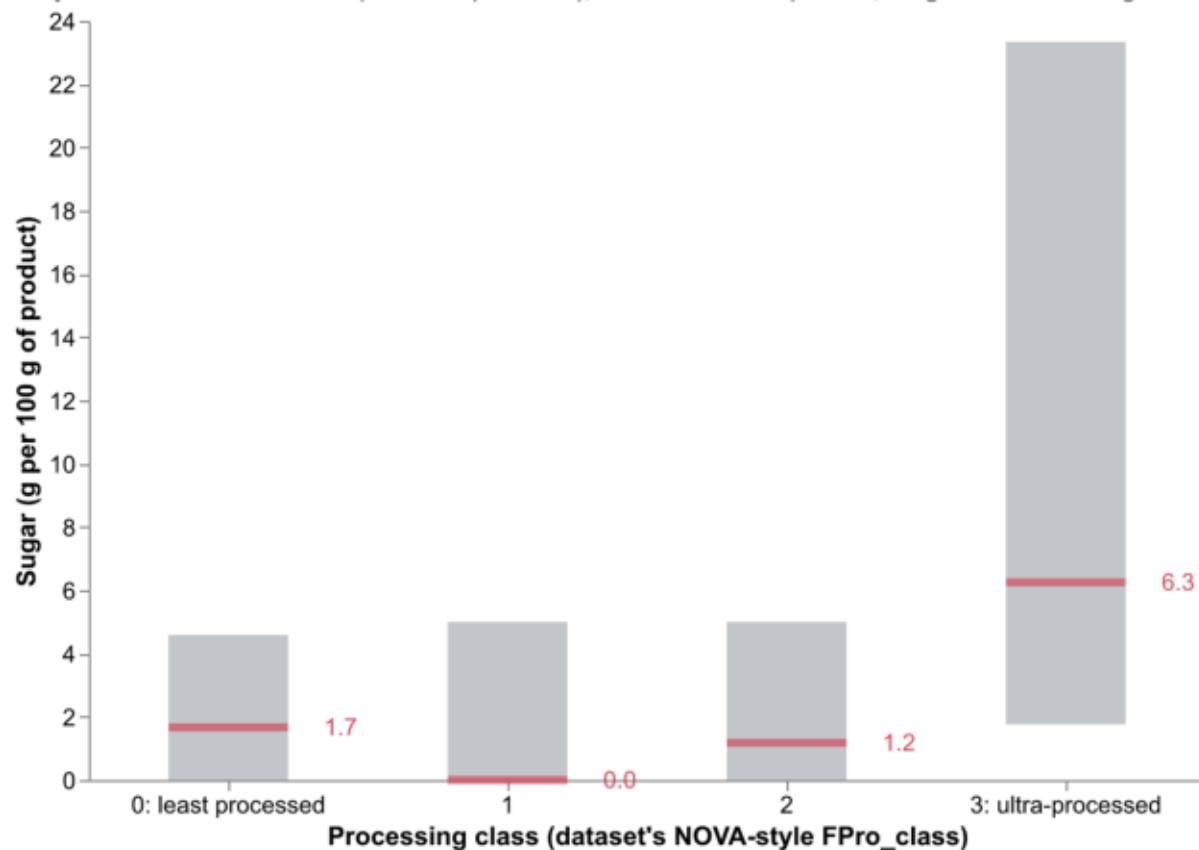


Food Processing Score (0 = least, 1 = most processed)

Plot C

Sugar jumps only in the ultra-processed class

Grey bar = middle 50% of items (25th-75th percentile); red tick = median (labeled; 6.3 g in class 3 vs 1.7 g or less elsewhere). Items with sugar > 100 g/100 g (data errors, 38 rows) removed.



Preferred plot

I prefer Plot A, the price-per-calorie scatter. It's the only one of the three that ties processing to something a shopper feels at the register, so it argues something instead of just describing a distribution. Both axes use position, the most accurate channel for comparing two quantities, and price is on a log scale because otherwise the cheap, heavily processed categories get squashed against the bottom. Plotting every product would be a smear, so each dot is a category median and its area shows how many items sit behind it. A 20-item category shouldn't look as solid as a huge one. Color only marks the categories I talk about, and the shaded box around cookies, cakes, bread, cereal and chips (about 30 cents per 100 kcal) is where I want people to look first. Plot B is easier to read, but the stores stock different mixes of food, so some of the gap could be assortment. Whole Foods does score lower than Target in 35 of the 50 categories all three stores carry, so mix isn't the whole story, but it's a caveat I'd have to state. Plot C is the weakest because sugar doesn't climb steadily across classes. Only class 3 stands out. Plot A has its own weakness: the trend is noisy (Spearman about -0.50 across categories) and a median hides the spread inside each category. In the final version I'll say that plainly and check whether item-level detail, or holding the store constant, makes the pattern clearer.